

AeroFit Treadmill Customer Analysis Report

1. Problem Statement

The objective is to identify characteristics of customers purchasing different AeroFit treadmills (KP281, KP481, KP781) and build customer profiles using descriptive analytics and probability analysis.

2. Dataset Overview

Dataset shape: 180 rows and 9 columns.

Column	Data Type
Product	category
Age	int64
Gender	category
Education	int64
MaritalStatus	category
Usage	int64
Fitness	int64
Income	int64
Miles	int64

3. Non■Graphical Analysis

Product	Count	Marginal Probability
KP281	80	0.444
KP481	60	0.333
KP781	40	0.222

Gender vs Product

Gender	KP281	KP481	KP781
Female	40	29	7
Male	40	31	33

Marital Status vs Product

Status	KP281	KP481	KP781
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Partnered	48	36	23
Single	32	24	17

Conditional Probability $P(\text{Product} \mid \text{Gender})$

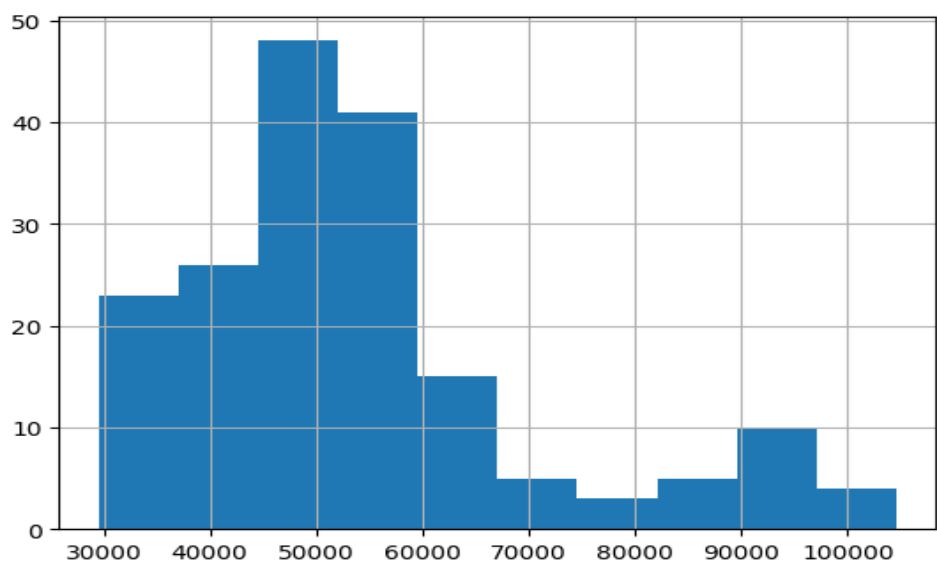
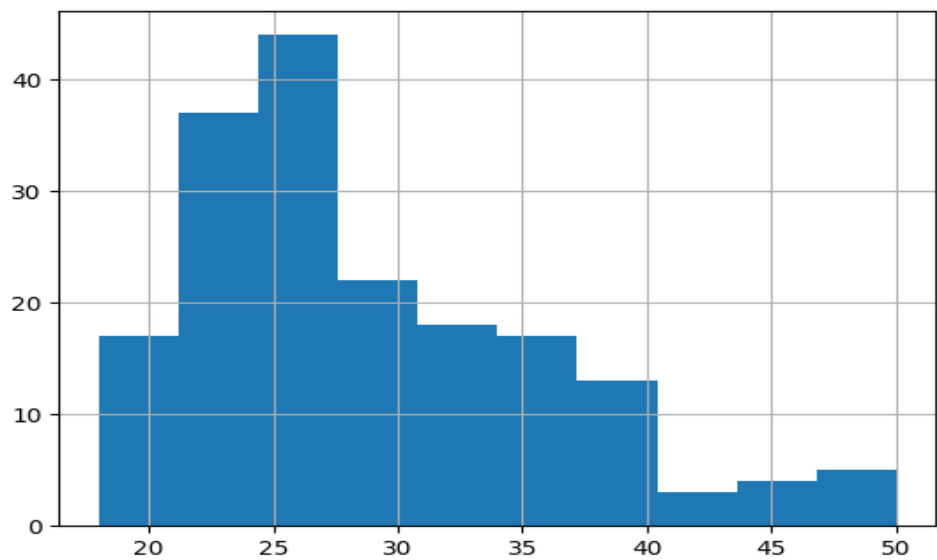
Gender	KP281	KP481	KP781
Female	0.526	0.382	0.092
Male	0.385	0.298	0.317

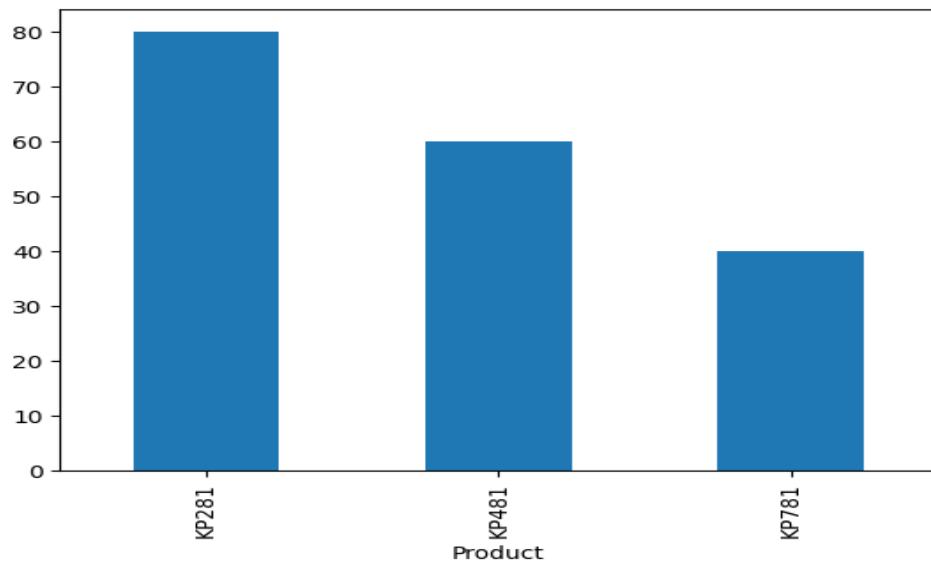
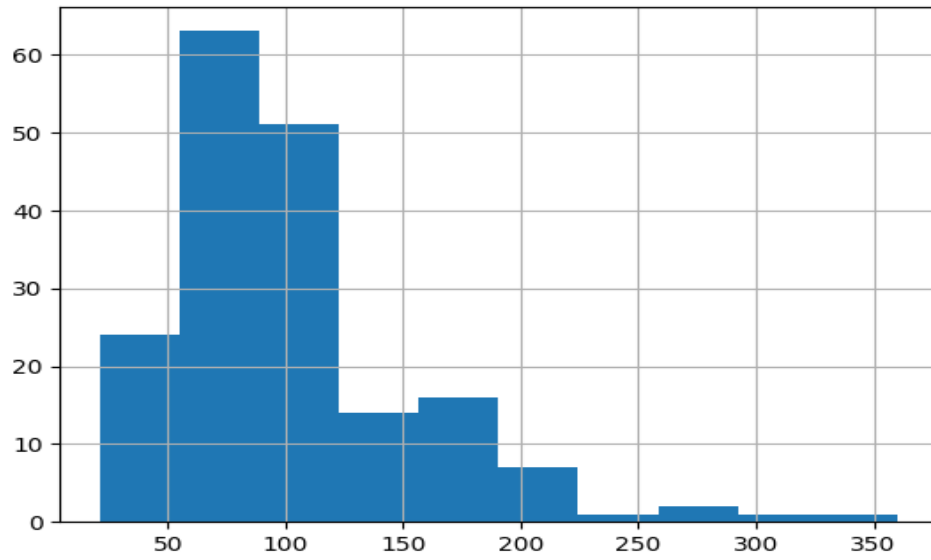
4. Missing Values & Outlier Detection

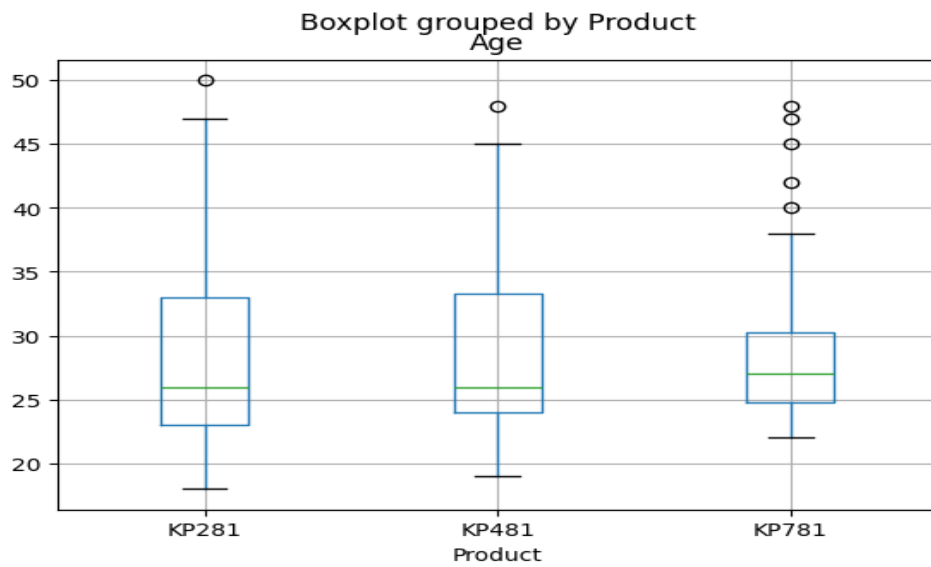
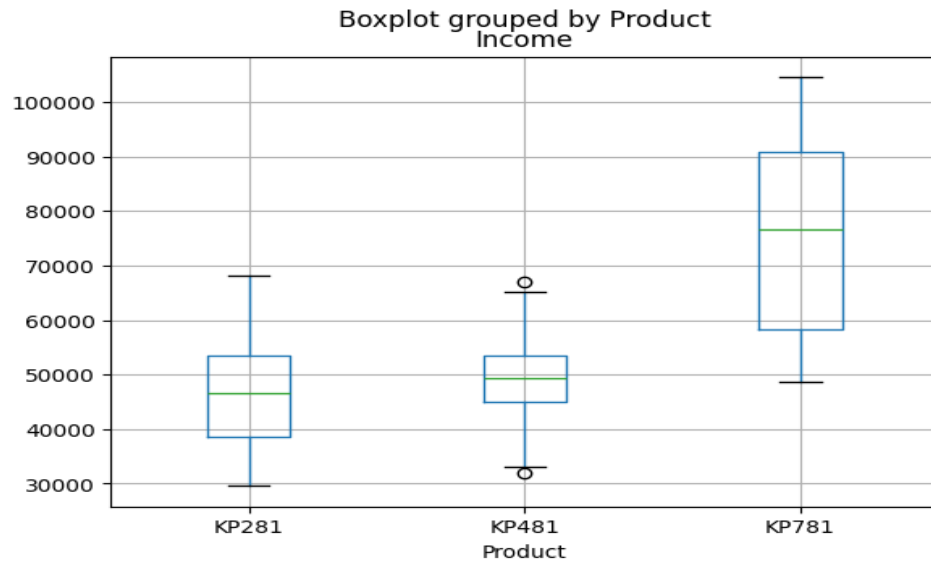
Column	Missing Values
Product	0
Age	0
Gender	0
Education	0
MaritalStatus	0
Usage	0
Fitness	0
Income	0
Miles	0

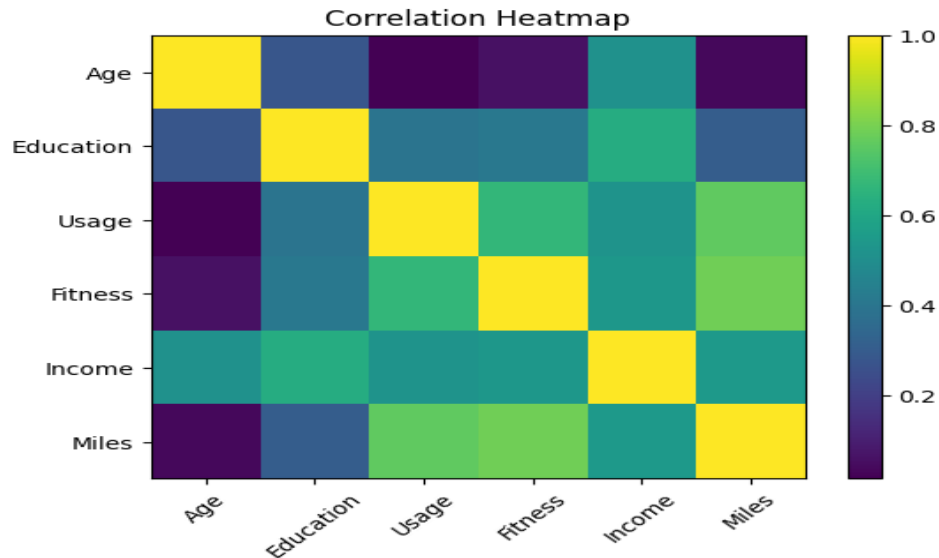
Column	Outlier Count
Age	5
Income	19
Miles	13
Usage	9
Fitness	2

5. Visual Analysis









6. Business Insights

- KP281 is the most frequently purchased treadmill, indicating high demand for entry-level fitness equipment.
- KP481 represents a mid-range segment targeting moderately active customers.
- KP781 buyers typically have higher income, higher fitness levels, and greater running mileage.
- Male customers appear more likely to purchase high-end treadmills compared with female customers.
- Customers with higher usage frequency tend to choose premium products.
- Income shows a positive relationship with advanced treadmill purchases.

7. Recommendations

- Promote KP281 to beginners and price-sensitive customers.
- Position KP481 as a performance upgrade for intermediate runners.
- Market KP781 to high-income fitness enthusiasts and serious runners.
- Offer bundle deals or EMI options to encourage upgrades from KP281 to KP481.
- Use targeted marketing campaigns based on income and fitness levels.
- Provide personalized recommendations in stores based on customer fitness and usage habits.